Hi-Fi Power Amplifier

Final Design, Protection and Measurement Report

Discrete Class-AB Stereo Power Amplifier
with STM32-Based Loudspeaker Protection

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Reference topology:	Rod Elliott Project 3A
PCB design:	KiCad
Protection MCU:	STM32G031K8
Final main PSU:	Two dedicated 500 VA transformers
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Engineering Construction, Commissioning and Verification Report

Contents

Abstract **v**

1 Project Overview and Final Configuration **1**

1.1 Project Objective 1

1.2 Final Hardware Architecture 1

1.3 System Signal and Power Paths 2

2 Power Amplifier Design and Operation **3**

2.1 Topology 3

2.2 Nominal Closed-Loop Gain 3

2.3 Class-AB Bias 3

2.4 Output Stage and Load Current 4

2.5 Output Stability Network 4

3 Main Power Supply, Grounding and Thermal Design **5**

3.1 Rectified Rail Voltage 5

3.2 Smoothing Capacitors and Ripple 5

3.3 Grounding Strategy 5

3.4 Thermal Management 6

4 STM32 Loudspeaker Protection Controller **7**

4.1 Functions 7

4.2 DC-Sense Interface 7

4.3 Temperature Monitoring 8

4.4 Relay Drivers 8

5 Construction and Commissioning **9**

5.1 Pre-Power Verification 9

5.2 Progressive Power-Up 9

5.3 Troubleshooting During Development 9

6 Measurement Equipment and Method 10

6.1 Laboratory Instruments 10

6.2 Measurement Conventions 10

7 Final Measurement Results 11

7.1 Transformer Secondary Measurements 11

7.2 DC Rails at Idle and Under an 8 Ohm Load 11

7.3 Rail Ripple Under Load 12

7.4 Quiescent Current and Thermal Stabilisation 12

7.5 DC Offset 12

7.6 Audio-Band Frequency Response and Gain 13

7.7 Upper -3 dB Bandwidth 14

7.8 Near-Clipping Output Power at 1 kHz 14

7.9 8 Ohm Loaded Output Operating Point 14

7.10 Protection Controller Measurements 15

8 Discussion of Results 16

8.1 Power-Supply Performance 16

8.2 Bias Behaviour 16

8.3 Near-Clipping Power and Load Dependence 16

8.4 Gain and Frequency Response 17

8.5 DC Offset and Speaker Safety 17

8.6 What the Available Data Do Not Establish 17

9 Troubleshooting and Engineering Corrections 18

9.1 Output-Stage Debugging 18

9.2 Protection-Input Debugging 18

9.3 Grounding and Load Diagnosis 18

10 Conclusion 19

A Instrumentation Record 20

B Design Source Files 21

C Original Measurement Log (Transcribed) 22

C.1 Transformer and Supply Measurements 22
C.2 Bias and DC Offset Log 23
C.3 Frequency Response and Loaded Test Log 23
C.4 Additional Handwritten Near-Clipping Test Log 23
C.5 Protection-Controller Log 23

List of Tables

1.1 Final system configuration. 2

6.1 Measurement instruments used for final verification. 10

7.1 Measured transformer secondary voltages. 11

7.2 Measured DC rail voltages. 11

7.3 Recorded rail ripple under 8 Ω load. 12

7.4 Measured bias voltage and calculated quiescent current. 12

7.5 Measured initial amplifier output DC offset. 13

7.6 Measured audio-band frequency response. 13

7.7 Near-clipping output voltages (measured) and sinusoidal power estimates (calculated), approximately 1 kHz. 14

7.8 Calculated quantities from the recorded 8 Ω loaded output test. 15

7.9 Measured protection-controller voltages. 15

C.1 Recorded transformer and supply voltages. 22

C.2 Original bias and DC observations. 23

C.3 Original controller voltage entries. 23

Abstract

This report documents the design, construction, protection system, commissioning and final available measurements of a stereo discrete Class-AB audio power amplifier derived from Rod Elliott's Project 3A. The completed system uses two identical amplifier channels, each supplied from its own 500 VA, 24–0–24 V_{RMS} transformer and local rectifier/filter section. A separate STM32G031K8-based controller supervises both amplifier outputs, two heatsink NTC sensors and the loudspeaker relays.

The final measurement record includes near-clipping output-voltage tests at 8 and 4 ohms, transformer secondary voltages, positive and negative DC rails at idle and under an 8 Ω load, supply ripple, cold and warm quiescent-current readings, DC offset, audio-band frequency response, upper −3 dB bandwidth, a defined 8 Ω loaded output operating point, and protection-controller voltages. The measured audio-band response from 20 Hz to 20 kHz remained at 22.6 V_{pp} for a recorded 1.2 V_{pp} input, corresponding to 18.83 V/V or 25.50 dB. The measured upper −3 dB frequency was approximately 250 kHz.

Additional handwritten near-clipping measurements at 1 kHz document output voltages into both 8-ohm and 4-ohm resistive loads. Output powers are calculated from these voltage readings, conditional on an approximately sinusoidal waveform just below visible clipping. THD-limited continuous output power, THD, SNR and damping factor were not independently measured. Where a numerical result is derived from a recorded measurement, it is explicitly identified as calculated rather than directly measured.

Chapter 1

Project Overview and Final Configuration

1.1 Project Objective

The objective of the project was to implement a complete stereo Hi-Fi power amplifier rather than only an isolated amplifier circuit. The final system integrates the analogue power stage, high-current power supply, thermal management, grounding, PCB construction and a dedicated loudspeaker-protection controller.

The analogue amplifier is based on Rod Elliott's Project 3A and uses a conventional discrete Class-AB topology with global negative feedback. The protection controller is physically and functionally separate from the audio signal path. It monitors DC conditions and temperature and controls normally-open loudspeaker relays.

1.2 Final Hardware Architecture

The final main-power arrangement is dual-mono at transformer level: one 500 VA, 24–0–24 V_{RMS} transformer is dedicated to the left channel and a second identical transformer is dedicated to the right channel. Each amplifier channel therefore has its own high-current AC source and rectified DC rails. The centre-tap/power-ground returns are brought to the system star-ground arrangement.

This final configuration supersedes the earlier single-transformer concept that appeared in preliminary design notes. The measured left and right transformer secondary values in Chapter 7 confirm the two-channel supply arrangement.

Table 1.1: Final system configuration.

Item	Final implementation
Amplifier	Discrete stereo Class-AB, Project 3A-derived
Main transformers	Two 500 VA transformers, one per channel
Secondary voltage	Nominal 24–0–24 V_{RMS} per transformer
Rectification	High-current bridge rectifier per channel supply
Filter capacitance	Two 4700 µF capacitors in parallel per rail, 9400 µF effective per rail
Output devices	MJL21193G (PNP) and MJL21194G (NPN)
Drivers	BD139 / BD140
Emitter resistors	0.33 Ω each (R13, R14)
Protection MCU	STM32G031K8 on NUCLEO-G031K8
Temperature sensing	Two 10 kΩ NTC thermistors
Speaker relays	Normally-open 12 V coils
Cooling	Passive heatsinks, no fan

1.3 System Signal and Power Paths

The audio path is

$$\text{Input} \rightarrow \text{Power Amplifier} \rightarrow \text{Protection Relay} \rightarrow \text{Loudspeaker}.$$

The main energy path is

$$\text{Mains} \rightarrow \text{Transformer} \rightarrow \text{Bridge Rectifier} \rightarrow \text{Filter Capacitors} \rightarrow \pm V_S \rightarrow \text{Amplifier}.$$

The protection controller operates in parallel with the audio path. It senses the slowly varying/DC component of each output and the two heatsink sensors, but it does not process the audio signal itself.

Chapter 2

Power Amplifier Design and Operation

2.1 Topology

Each channel contains an input network, differential input stage, voltage-amplification stage, bias/driver circuitry and a complementary high-current output stage. Global negative feedback returns a fraction of the output to the inverting input so that the amplifier corrects the difference between the requested and actual output:

$$V_{\text{error}} = V_{\text{input}} - V_{\text{feedback}}.$$

The differential stage provides the error signal, the voltage-amplification stage develops the required internal voltage swing, and the driver/output devices provide the load current required by the loudspeaker.

2.2 Nominal Closed-Loop Gain

The nominal feedback network is approximately

$$R_5 = 22\,\text{k}\Omega, \qquad R_4 = 1\,\text{k}\Omega.$$

The ideal non-inverting closed-loop gain is therefore

$$A_v \approx 1 + \frac{R_5}{R_4} = 23,$$

or

$$A_{v,\text{dB}} = 20\log_{10}(23) \approx 27.2\,\text{dB}.$$

This is a design value, not a substitute for measured gain. The final measured values are reported separately in Chapter 7.

2.3 Class-AB Bias

The two emitter resistors are

$$R_{13} = R_{14} = 0.33\,\Omega.$$

When the voltage is measured across both resistors together,

$$R_{13} + R_{14} = 0.66\,\Omega,$$

and the corresponding quiescent current is calculated as

$$I_Q = \frac{V_{R13+R14}}{0.66\,\Omega}.$$

A 50 mA nominal reference would correspond to approximately

$$V_{R13+R14} = 33\,\text{mV}.$$

The actual final cold and warm values were higher than this nominal reference and are reported without alteration in Section 7.4.

2.4 Output Stage and Load Current

The final output pair is MJL21193G (PNP) and MJL21194G (NPN), manufactured by onsemi. The output devices are driven by BD139/BD140 transistors and are heatsink mounted.

For a resistive load,

$$I_{\text{RMS}} = \frac{V_{\text{RMS}}}{R_L}, \qquad P_{\text{out}} = \frac{V_{\text{RMS}}^2}{R_L}.$$

A 4 Ω load requires twice the current of an 8 Ω load at the same output voltage, so it is a more severe test of the transistors, emitter resistors, PCB copper, supply and thermal system. A subsequent handwritten test sheet records near-clipping voltage readings for both 8 Ω and 4 Ω loads. Corresponding calculated power values appear in Section 7.8.

2.5 Output Stability Network

The output includes an RC Zobel/Boucherot network to present a more predictable high-frequency load. This is important because a loudspeaker and its cable are reactive rather than purely resistive, and the global-feedback amplifier must remain stable when driving practical loads.

Chapter 3

Main Power Supply, Grounding and Thermal Design

3.1 Rectified Rail Voltage

For a nominal $24\,\mathrm{V_{RMS}}$ half-secondary,

$$V_{\mathrm{peak}} = 24\sqrt{2} \approx 33.94\,\mathrm{V}.$$

The real no-load value can be slightly higher because of transformer regulation, while loaded rails fall because of winding resistance, rectifier loss, capacitor ESR, wiring and capacitor discharge between charging peaks.

The final recorded idle rails were approximately ±34.5 V, in good agreement with the expected range for the selected transformer voltage.

3.2 Smoothing Capacitors and Ripple

Two 4700 µF capacitors are paralleled on each rail, giving

$$C_{\mathrm{rail}} = 9400\,\mu\mathrm{F}.$$

At 50 Hz mains frequency, full-wave rectification produces a principal charging/ripple frequency of approximately 100 Hz. A first-order estimate is

$$\Delta V \approx \frac{I}{fC},$$

although the real waveform depends on the pulsed charging current and the audio load.

3.3 Grounding Strategy

The amplifier contains both low-level signal references and high-current return paths. The design therefore distinguishes signal ground, amplifier power ground, loudspeaker return, rectifier/filter return, protection-controller ground and protective earth.

High-current loudspeaker and capacitor-charging currents are returned to the main power-ground/star region instead of being allowed to flow through the input reference path. The metal chassis is bonded to protective earth independently of any audio-ground strategy.

3.4 Thermal Management

The output transistors are mounted to external passive heatsinks with appropriate electrical insulation and thermal interface material. The relevant thermal path can be expressed as

$$T_J = T_A + P_D \left(R_{\theta JC} + R_{\theta CS} + R_{\theta SA} \right).$$

The warm quiescent-current data show a clear increase relative to the cold condition, which is why final bias should always be assessed after thermal stabilisation rather than only at switch-on.

Chapter 4

STM32 Loudspeaker Protection Controller

4.1 Functions

The STM32G031K8-based controller provides the following functions:

- approximately 3 s startup delay;
- left and right DC-output monitoring;
- two NTC temperature channels;
- sensor-plausibility/fault detection;
- relay control for left and right loudspeaker outputs;
- status/fault LED indication.

The normally-open relay arrangement is fail-safe with respect to loss of relay drive: if controller power or GPIO drive disappears, the loudspeaker is disconnected.

4.2 DC-Sense Interface

The power-amplifier output cannot be connected directly to a 0 V–3.3 V ADC input. The implemented sensing network attenuates and biases the amplifier output. Its approximate static relationship is

$$V_{\mathrm{ADC}} \approx 1.597 + 0.03226 V_{\mathrm{AMP}},$$

where V_{AMP} is the slowly varying/DC component of the amplifier output.

Therefore a healthy amplifier output close to 0 V should appear near 1.6 V at the ADC, while positive or negative persistent DC shifts the ADC voltage above or below this centre value.

The sensing network also includes analogue low-pass filtering. This is essential because the controller should respond to persistent DC rather than the instantaneous audio waveform. The final measurement notes record that signals below approximately 15 Hz are treated by the protection system as DC-like input. This system behaviour limited the possibility of measuring a lower -3 dB cutoff below that region.

4.3 Temperature Monitoring

Each heatsink uses a 10 kΩ NTC thermistor. The resistance decreases with increasing temperature and can be approximated by

$$R(T) = R_{25} \exp\left[B\left(\frac{1}{T} - \frac{1}{T_{25}}\right)\right].$$

The firmware uses an approximate 100 °C over-temperature trip threshold. Sensor values outside the configured plausible range are also treated as faults so that a disconnected thermistor cannot be interpreted as a safe temperature.

4.4 Relay Drivers

Each 12 V relay coil is driven by a BC337 low-side transistor. The STM32 GPIO therefore supplies only the transistor base current, not the relay-coil current. Flyback diodes across the coils absorb the inductive current when the drivers switch off.

The final measured relay-driver and coil voltages are presented in Section 7.10.

Chapter 5

Construction and Commissioning

5.1 Pre-Power Verification

Before first power-up, polarity-sensitive components and transistor pinouts were checked, the supply rails were checked for shorts to ground, heatsink insulation was verified, and the speaker output was kept disconnected. The bridge rectifier and electrolytic capacitor polarities were treated as critical checks because an error in this area can cause immediate high-current failure.

5.2 Progressive Power-Up

Commissioning was performed progressively: supply verification, no-load amplifier operation, bias adjustment, DC-offset checks, protection-board testing and finally loaded testing. This approach reduces the chance that a single wiring or component error can damage the output stage or loudspeaker.

5.3 Troubleshooting During Development

The construction required several debugging iterations involving output-stage connections, driver bias, fuse operation, floating protection inputs and grounding. The final working configuration was obtained by verifying transistor health, checking DC operating points, grounding the audio input during diagnosis, correcting protection behaviour for floating DC-sense inputs and rechecking the amplifier after each hardware change.

The troubleshooting process is part of the engineering result: successful operation was not inferred only from schematic correctness or firmware compilation but verified on the assembled hardware.

Chapter 6

Measurement Equipment and Method

6.1 Laboratory Instruments

The final measurement campaign used the following main instruments:

Table 6.1: Measurement instruments used for final verification.

Instrument	Model	Main use in this project
Oscilloscope	UNI-T UPO1202	Output waveform, peak-to-peak voltage, frequency response, ripple and clipping observation
Signal generator	OWON DGE1030	Sine-wave excitation for gain and frequency-response tests
Digital multimeter	OWON XDM2041	DC rails, offset, bias voltage, ADC/relay voltages and general commissioning

6.2 Measurement Conventions

Peak-to-peak and RMS quantities are not interchangeable. For a sinusoidal waveform,

$$V_{\mathrm{RMS}} = \frac{V_{pp}}{2\sqrt{2}}.$$

Voltage gain is calculated using like-for-like amplitude definitions:

$$A_v = \frac{V_{out}}{V_{in}}, \qquad A_{v,\mathrm{dB}} = 20\log_{10}(A_v).$$

When output power is derived from a recorded sinusoidal voltage across a resistive load,

$$P_{out} = \frac{V_{\mathrm{RMS}}^2}{R_L}.$$

Derived values in this report are calculated only from recorded measurements; they are not additional measurements.

Chapter 7

Final Measurement Results

7.1 Transformer Secondary Measurements

The two dedicated transformers were measured from each outer secondary lead to the centre tap.

Table 7.1: Measured transformer secondary voltages.

Measurement	Left channel	Right channel
AC1 to centre tap	25.652 V_{RMS}	25.628 V_{RMS}
AC2 to centre tap	25.631 V_{RMS}	25.516 V_{RMS}
Calculated outer-to-outer sum	51.283 V_{RMS}	51.144 V_{RMS}

The outer-to-outer values above are calculated by summing the two measured half-secondary magnitudes; they were not separately recorded as direct measurements.

7.2 DC Rails at Idle and Under an 8 Ohm Load

Table 7.2: Measured DC rail voltages.

Condition	Left channel	Right channel
Positive rail, idle	34.521 V	34.566 V
Negative rail, idle	−34.532 V	−34.522 V
Positive rail, 8 Ω load	33.534 V	33.678 V
Negative rail, 8 Ω load	−33.560 V	−33.701 V

The calculated positive-rail sag was 0.987 V (2.86 %) for the left channel and 0.888 V (2.57 %) for the right channel. The negative-rail magnitude decreased by 0.972 V (2.81 %) on the left and 0.821 V (2.38 %) on the right.

7.3 Rail Ripple Under Load

The recorded ripple values under the 8 Ω load condition were:

Table 7.3: Recorded rail ripple under 8 Ω load.

Rail	Left channel	Right channel
Positive rail	1.5 V	1.4 V
Negative rail	1.6 V	1.4 V

The original measurement sheet records only the ripple magnitude in volts and does not separately state whether the displayed oscilloscope quantity was peak-to-peak or another amplitude statistic. The values are therefore reproduced exactly without relabelling them as V_{pp} or V_{RMS}.

7.4 Quiescent Current and Thermal Stabilisation

The measured voltage across R13+R14 was converted to quiescent current using

$$I_Q = \frac{V_{R13+R14}}{0.66\,\Omega}.$$

Table 7.4: Measured bias voltage and calculated quiescent current.

	Cold		Warm / stabilised	
Channel	$V_{R13+R14}$	I_Q	$V_{R13+R14}$	I_Q
Left	41.700 mV	63.18 mA	52.55 mV	79.62 mA
Right	41.730 mV	63.23 mA	55.71 mV	84.41 mA

The warm current increased by approximately 26.0 % on the left channel and 33.5 % on the right relative to the cold readings. Both final warm values are above the earlier 50 mA nominal reference. Because no additional adjustment measurement is available, the report records the stabilised values as built rather than replacing them with the nominal design target.

7.5 DC Offset

The initial DC-offset measurements were:

Table 7.5: Measured initial amplifier output DC offset.

Channel	Initial DC offset
Left	−54.54 mV
Right	−28.203 mV

These are small compared with the supply rails and do not represent a dangerous loudspeaker DC condition.

A later no-signal measurement in the same record produced 42.300 mV on the left channel and −23.764 mV on the right. Because the two measurement sets were taken under different undocumented timing/operating conditions, they are treated as separate snapshots rather than averaged together.

7.6 Audio-Band Frequency Response and Gain

The recorded frequency-response test used

$$V_{in} = 1.2\,\mathrm{V_{pp}}.$$

At every recorded frequency from 20 Hz through 20 kHz, both channels produced 22.6 $\mathrm{V_{pp}}$.

Table 7.6: Measured audio-band frequency response.

Frequency	Left output	Right output
20 Hz	22.6 $\mathrm{V_{pp}}$	22.6 $\mathrm{V_{pp}}$
50 Hz	22.6 $\mathrm{V_{pp}}$	22.6 $\mathrm{V_{pp}}$
100 Hz	22.6 $\mathrm{V_{pp}}$	22.6 $\mathrm{V_{pp}}$
500 Hz	22.6 $\mathrm{V_{pp}}$	22.6 $\mathrm{V_{pp}}$
1 kHz	22.6 $\mathrm{V_{pp}}$	22.6 $\mathrm{V_{pp}}$
5 kHz	22.6 $\mathrm{V_{pp}}$	22.6 $\mathrm{V_{pp}}$
10 kHz	22.6 $\mathrm{V_{pp}}$	22.6 $\mathrm{V_{pp}}$
20 kHz	22.6 $\mathrm{V_{pp}}$	22.6 $\mathrm{V_{pp}}$

The measured voltage gain over this sweep is therefore

$$A_v = \frac{22.6}{1.2} = 18.83\ \mathrm{V/V},$$

which corresponds to

$$A_{v,\mathrm{dB}} = 20\log_{10}(18.83) = 25.50\,\mathrm{dB}.$$

The response is flat to the resolution of the recorded data across the complete 20 Hz–20 kHz audio band. This is qualitatively consistent with the wideband audio response intended by the Rod Elliott Project 3A reference design; it is not a digitisation of Elliott's published frequency-response curve. The observed gain must be distinguished from the nominal feedback-network estimate.

7.7 Upper -3 dB Bandwidth

The measured upper $-3\,\mathrm{dB}$ frequency was approximately

$$f_H = 250\,\mathrm{kHz}.$$

For the $22.6\,\mathrm{V_{pp}}$ mid-band reference, the $-3\,\mathrm{dB}$ voltage level is

$$V_{-3\,\mathrm{dB}} = 0.707 \times 22.6 \approx 15.98\,\mathrm{V_{pp}}.$$

The lower $-3\,\mathrm{dB}$ cutoff was not established. The system-level measurement was limited by the protection controller, which interprets frequencies below approximately 15 Hz as DC-like input. However, the measured 20 Hz point showed no attenuation relative to the rest of the recorded audio-band sweep.

7.8 Near-Clipping Output Power at 1 kHz

A later handwritten worksheet records output voltages at approximately 1 kHz with non-inductive resistive loads, adjusted to just below visible clipping. The voltage measurements are directly recorded; RMS voltage and output power are calculated assuming a sinusoidal waveform:

$$V_{\mathrm{RMS}} = \frac{V_{pp}}{2\sqrt{2}}, \qquad P_{\mathrm{out}} = \frac{V_{pp}^2}{8R_L}. \tag{7.1}$$

Table 7.7: Near-clipping output voltages (measured) and sinusoidal power estimates (calculated), approximately 1 kHz.

Channel	Load	Input (V_{pp})	Output (V_{pp})	V_{RMS}	Power (W)
Left	$8\,\Omega$	2.72	62.00	21.92	60.06
Right	$8\,\Omega$	2.84	61.20	21.64	58.52
Left	$4\,\Omega$	2.48	58.80	20.79	108.05
Right	$4\,\Omega$	2.52	58.40	20.65	106.58

The readings correspond to approximately 60 W per channel into $8\,\Omega$ and 107 W per channel into $4\,\Omega$. They are near-clipping estimates, not certified continuous or distortion-limited ratings. The worksheet does not provide THD, exact clipping margin, test duration or temperature at the output-power points. Each channel was measured individually; these values do not establish simultaneous stereo full-power operation.

7.9 8 Ohm Loaded Output Operating Point

With an $8\,\Omega$ load and $1.2\,\mathrm{V_{pp}}$ input, the recorded outputs were $27.2\,\mathrm{V_{pp}}$ on the left channel and $26.6\,\mathrm{V_{pp}}$ on the right.

Table 7.8: Calculated quantities from the recorded 8 Ω loaded output test.

Quantity	Left	Right
Measured output	27.2 V_{pp}	26.6 V_{pp}
Calculated gain	22.67 V/V	22.17 V/V
Calculated gain	27.11 dB	26.91 dB
Calculated output	9.617 V_{RMS}	9.405 V_{RMS}
Calculated power into 8 Ω	11.56 W	11.06 W

The raw record lists 27.23 dB for the right-channel loaded gain. Recalculation from its recorded 26.6 V_{pp} output and 1.2 V_{pp} input gives 26.91 dB; the recalculated value is used here while the original entry is transcribed in Appendix C.

This was a defined loaded operating point, not a recorded maximum-unclipped-output test. Consequently, 11.56 W and 11.06 W are powers at the measured test point and must not be interpreted as the maximum capability of the amplifier.

7.10 Protection Controller Measurements

Table 7.9: Measured protection-controller voltages.

Measurement	Left / Channel 1	Right / Channel 2
Healthy DC-sense ADC voltage	1.6014 V	1.5969 V
Relay-driver base voltage, ON	3.2932 V	3.2931 V
Relay-coil voltage, ON	11.723 V	11.735 V
NTC channel at room temperature	1.1230 V	1.1225 V

The healthy ADC readings are centred very close to the predicted approximately 1.6 V value of the DC-sense network. Relay-driver base voltages are essentially the STM32 3.3 V logic level, and the relay coils receive approximately 11.7 V in the energised state.

The measurement record states that all other protection-system tests passed. This includes the functional fault and relay behaviour exercised during commissioning. Detailed numerical timing and trip-point traces were not separately preserved, so only the pass result is claimed here.

Chapter 8

Discussion of Results

8.1 Power-Supply Performance

The measured transformer half-secondary values are around 25.5 V_{RMS}–25.7 V_{RMS}, slightly above the nominal 24 V_{RMS} rating as expected under light loading. The resulting idle DC rails of approximately ±34.5 V are consistent with this secondary voltage after rectification and filtering.

Under the recorded 8 Ω load test, the rail magnitude fell only about 2.4 %–2.9 %. The left/right and positive/negative rails also remained closely symmetrical. This indicates good supply stiffness at the specific test operating point.

8.2 Bias Behaviour

The cold quiescent current was about 63 mA on both channels, showing good channel-to-channel agreement immediately after startup. The warm values increased to approximately 80 mA and 84 mA. The increase confirms the importance of thermal stabilisation when setting a Class-AB bias network.

The report does not claim that these warm currents are an optimum universal bias target; they are simply the final measured stabilised values. No later adjustment measurement is available.

8.3 Near-Clipping Power and Load Dependence

The measured near-clipping output swing is lower with a 4 Ω load than with an 8 Ω load, as expected when the output stage and supply deliver more current. The corresponding calculated powers are approximately 59–60 W into 8 Ω and 107–108 W into 4 Ω, depending on channel. These are conditional sine-wave estimates, not a THD-certified rated-power specification.

8.4 Gain and Frequency Response

The audio-band sweep showed no recorded change in output amplitude between 20 Hz and 20 kHz. The measured sweep gain of 25.50 dB is somewhat below the simple nominal feedback-network estimate of about 27.2 dB. The separate loaded 8 Ω operating point gave gains around 27 dB. Because the test conditions were not identical, these results are presented as separate measurement sets rather than forced into a single number.

The upper −3 dB point at approximately 250 kHz is far above the audio band and shows substantial high-frequency bandwidth. The lower cutoff cannot be claimed because the protection system prevents a clean system-level test below approximately 15 Hz; nevertheless, the 20 Hz result remained at the mid-band level.

8.5 DC Offset and Speaker Safety

All recorded normal-operation output offsets are in the tens of millivolts, far below the dangerous multi-volt DC conditions the protection controller is designed to detect. The measured DC-sense ADC centres also agree closely with the analytical approximately 1.597 V healthy value, supporting correct biasing of the protection inputs.

8.6 What the Available Data Do Not Establish

Because no further measurements can be performed, the following quantities are intentionally not reported as measured performance specifications:

- THD-limited continuous output power at a defined distortion threshold;
- simultaneous two-channel maximum output under sustained load;
- total harmonic distortion (THD or THD+N);
- signal-to-noise ratio and residual noise floor;
- damping factor / output impedance;
- slew rate;
- channel crosstalk;
- exact lower −3 dB cutoff;
- detailed thermal trip curve versus heatsink temperature.

This limitation does not invalidate the measurements that were obtained; it only defines the boundary of what can be claimed from the available evidence.

Chapter 9

Troubleshooting and Engineering Corrections

9.1 Output-Stage Debugging

During commissioning, one amplifier channel required extensive diagnosis of driver/output-stage connections, fuse operation and DC operating points. Output transistors and drivers were checked independently, and the amplifier was repeatedly tested with the input grounded so that a floating input could not be mistaken for an internal amplifier fault.

The final hardware operated correctly after the relevant connection and bias issues were resolved. This experience reinforced the importance of checking the exact pinout of each semiconductor against its datasheet rather than relying on package appearance.

9.2 Protection-Input Debugging

Floating protection-controller DC-sense inputs initially produced an apparent DC-fault condition because the ADC nodes sat near their internally biased centre even when no amplifier source was connected. Firmware/logic handling was adjusted so that the practical system no longer produced the false fault observed during commissioning.

9.3 Grounding and Load Diagnosis

A hum/oscillation symptom was observed with one particular $4\,\Omega$ loudspeaker but not with an $8\,\Omega$ resistor or other speakers. This indicated that the symptom was load/system-specific rather than a universal no-load amplifier failure. High-current grounding, speaker wiring and reactive-load behaviour were therefore treated separately from basic DC stability.

Chapter 10

Conclusion

The project resulted in a complete working stereo Class-AB amplifier system with independent high-current transformers for the two channels and an STM32-based loudspeaker-protection controller. The final system architecture addresses not only voltage gain but also power delivery, grounding, thermal behaviour, startup sequencing and fault isolation.

The recorded main-supply measurements show approximately ±34.5 V idle rails with only a few percent sag at the documented 8 Ω operating point. Quiescent-current measurements are closely matched when cold and rise after warm-up to approximately 79.6 mA and 84.4 mA. Normal output DC offsets remained in the tens of millivolts.

The measured 20 Hz–20 kHz sweep was flat at 22.6 V_{pp} output for 1.2 V_{pp} input, corresponding to 25.50 dB gain under that measurement condition. The upper −3 dB point was approximately 250 kHz. A separate 8 Ω loaded test at 1.2 V_{pp} input produced 27.2 V_{pp} left and 26.6 V_{pp} right, corresponding to calculated powers of approximately 11.56 W and 11.06 W at that specific operating point.

The protection-controller ADC, relay-driver, relay-coil and NTC readings were all consistent with normal operation, and the remaining protection tests were recorded as passed. These results verify the functional operation of the assembled amplifier and protection system within the scope of the measurements that were actually captured.

The later handwritten near-clipping test gives calculated sinusoidal power estimates of 60.06/58.52 W (left/right) into 8 ohms and 108.05/106.58 W into 4 ohms. No distortion-limited continuous-power, distortion or noise specification is invented in this final report. The documented measurements are the complete final measurement set available for this build.

Appendix A

Instrumentation Record

The principal instruments used to obtain the final results were:

- **UNI-T UPO1202** digital oscilloscope, used for waveform, amplitude, frequency-response and ripple observation;
- **OWON DGE1030** signal generator, used as the sine-wave source;
- **OWON XDM2041** digital multimeter, used for DC rails, bias, offset and protection-controller voltage measurements.

These model numbers are included so that the measurement environment can be reproduced or compared against future verification work.

Appendix B

Design Source Files

The final project source set associated with this report includes KiCad design files for the power amplifier, power supply and related system boards. The design authority for component connections and PCB implementation remains the KiCad project files, while this report documents system operation and the final available verification data.

Relevant supplied source filenames include:

- `power_amplifier.kicad_sch`;
- `Power Amplifier Rod Elliot Project 3A.kicad_pcb`;
- `Power Amplifier Rod Elliot Project 3A.kicad_sch`;
- `power_supply.kicad_sch`.

Appendix C

Original Measurement Log (Transcribed)

This appendix provides a readable transcription of the values supplied in the original three-page measurement sheet. Units, channel assignments, and test conditions are preserved; the derived and corrected values are discussed in Chapter 7. The original worksheet is supplied separately in the source package, not embedded as an image.

C.1 Transformer and Supply Measurements

Table C.1: Recorded transformer and supply voltages.

Measurement	Left	Right
Transformer AC1 to centre tap (V RMS)	25.652	25.628
Transformer AC2 to centre tap (V RMS)	25.631	25.516
Positive rail at idle (V DC)	34.521	34.566
Negative rail at idle (V DC)	-34.532	-34.522
Positive rail, 8-ohm load (V DC)	33.534	33.678
Negative rail, 8-ohm load (V DC)	-33.560	-33.701
Positive rail ripple, 8-ohm load (V, recorded)	1.5	1.4
Negative rail ripple, 8-ohm load (V, recorded)	1.6	1.4

C.2 Bias and DC Offset Log

Table C.2: Original bias and DC observations.

Measurement	Left	Right
R13 + R14 cold (mV)	41.700	41.730
R13 + R14 warm/stabilised (mV)	52.55	55.71
Initial DC offset (mV)	-54.54	-28.203
Output with no signal input (mV)	42.300	-23.764

C.3 Frequency Response and Loaded Test Log

The source log specifies an input of $1.2\,V_{pp}$ and the following eight frequency points, at each of which both outputs were reported as $22.6\,V_{pp}$: 20, 50, 100, 500, 1000, 5000, 10000, and 20 000 Hz. It reports an upper -3 dB point of 250 kHz and notes that the protection system interprets input frequencies below about 15 Hz as DC.

With an $8\,\Omega$ load and a recorded $1.2\,V_{pp}$ input, the log states $27.2\,V_{pp}$ (left) and $26.6\,V_{pp}$ (right). The original recorded gain entries were 27.11 dB and 27.23 dB, respectively. The right-channel gain is recomputed in the main text from the recorded input/output voltages; the original entry is preserved here as written.

C.4 Additional Handwritten Near-Clipping Test Log

The additional photographed worksheet records, at approximately 1 kHz: right channel, $8\,\Omega$, input $2.84\,V_{pp}$, output $61.20\,V_{pp}$; left channel, $8\,\Omega$, input $2.72\,V_{pp}$, output $62.00\,V_{pp}$; right channel, $4\,\Omega$, input $2.52\,V_{pp}$, output $58.40\,V_{pp}$; and left channel, $4\,\Omega$, input $2.48\,V_{pp}$, output $58.80\,V_{pp}$. These measurements were not part of the original three-page typed log. Calculations are given in Section 7.8.

C.5 Protection-Controller Log

Table C.3: Original controller voltage entries.

Measurement	Left	Right
Healthy DC-sense ADC input (V)	1.6014	1.5969
Relay-driver base, relay ON (V)	3.2932	3.2931
Relay coil, relay ON (V)	11.723	11.735
NTC channel voltage at room temperature (V)	1.1230	1.1225

The original sheet states: “ALL OTHER PROTECTION SYSTEM TEST ARE PASSED.” No individual measurements, event timestamps, or waveforms were supplied for those additional tests. This summary is therefore retained as an operator-reported result rather than independently quantified evidence.

Bibliography

[1] Rod Elliott, *Project 3A - 60-100 W Power Amplifier*, Elliott Sound Products.

[2] STMicroelectronics, *STM32G031K8 Datasheet and Reference Documentation.*

[3] onsemi, *MJL21193G PNP Audio Power Transistor Datasheet.*

[4] onsemi, *MJL21194G NPN Audio Power Transistor Datasheet.*